

Carl A. Schmidt

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Appointments

Research Assistant Professor, Boston University	2021 - Present
Research Scientist, Boston University	2017 - 2021
Research Associate, Sorbonne Univ. CNRS, Paris (F. Leblanc Supervisor)	2015 - 2017
Research Associate, Univ. of Virginia (R. E. Johnson Supervisor)	2013 - 2015

Expertise: Planetary aeronomy, Monte-Carlo modelling, telescope-based spectroscopy & instrumentation, aurorae & plasmas

Education

Ph.D., Astronomy, Boston University	2013
Thesis: <i>Mercury's Sodium Exosphere</i> (M. Mendillo Advisor)	
B.A., Physics, University of Colorado	2005

Advising

Post-Doctoral and Research Scientists:

- Yudai Suzuki (Fall 2025 -) JSPS Postdoctoral Fellow
- Candace Gray (Fall 2025 -) Research Scientist
- Quanzhi Ye (Fall 2022 -) Visiting Researcher (Joint appt w/ Univ. of Maryland)

Graduate Research Advisor:

- Patrick Lierle (Fall 2022 -)
- Emma Lovett (Fall 2021 -) NASA FINESST Graduate Fellow

Undergraduate Research Advisor (6 students since 2018)

Selected Peer-Reviewed Articles (from 57 total, students underlined)

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- **C. Schmidt**, L. Roth, S. Mendenhall, K. Retherford, J. Spencer (2026) *A Lack of Carbon at Io and its Plasma Torus*, Icarus, V454, 117088, [DOI](#).

- P. R. Lierle, E. Lovett, **C. Schmidt**, A. Merkel (2025) *Characterizing Dynamical Processes in Surface-Bound Exospheres Via Resolved Sodium D Emissions*, *Frontiers in Astronomy and Space Sciences*, V12, 1585683, [DOI](#).
- E. Lovett, **C. Schmidt**, P. R. Lierle (2025) *Europa's Sodium and Potassium Exosphere during Juno's Flyby*, *The Planetary Science Journal*, V6, 178, [DOI](#).
- P. R. Lierle & **C. Schmidt** (2025) *Sodium and Potassium Linewidths as an Atmospheric Escape Diagnostic at Mercury*, *Journal of Geophysical Research: Planets*, V130 (7), e2025JE008975, [DOI](#).
- **C. Schmidt** (2024) *On the Implications of Ground-Based High-Definition Imaging of Io's Surface*. *Geophysical Research Letters*, V51, 21, e2024GL112169, [DOI](#).
- P. Lierle, **C. Schmidt**, J. Baumgardner, L. Moore and E. Lovett (2023) *The Rapid Imaging Planetary Spectrograph*. *Publications of the Astronomical Society of the Pacific*, 135, 095002, [arXiv](#), [DOI](#).
- **C. Schmidt**, M. Sharov, K. de Kleer, N. Schneider, I. de Pater, P.H. Phipps, A. Conrad, L. Moore, P. Withers, J. Spencer, J. Morgenthaler, I. Ilyin, K. Strassmeier, C. Veillet, J. Hill, and M. Brown (2023) *Io's Optical Aurorae in Jupiter's Shadow*. *The Planetary Science Journal*, V4, 36, [DOI](#).
- P. R. Lierle, **C. Schmidt**, J. Baumgardner, L. Moore, T. Bida and R. Swindle (2022) *The Spatial Distribution and Temperature of Mercury's Potassium Exosphere*. *The Planetary Science Journal*, V3, 87, [DOI](#).
- **C. Schmidt** (2022) *Doppler-Shifted Alkali D Absorption as Indirect Evidence for Exomoons*. *Frontiers in Astronomy and Space Sciences*, Vol. 9, 801873, [DOI](#).
- **C. Schmidt**, J. Baumgardner, L. Moore, T. A. Bida, R. Swindle and P. Lierle (2020) *The Rapid Imaging Planetary Spectrograph: Observations of Mercury's Sodium Exosphere in Twilight*. *The Planetary Science Journal*, V1, 4, [DOI](#).
- **C. Schmidt**, N. Schneider, F. Leblanc, C. Gray, J. Morgenthaler, J. Turner and C. Grava (2018) *A Survey of Visible S⁺ Emission in Io's Plasma Torus During the Hisaki Epoch*. *Journal of Geophysical Research: Space Physics*, 123, 7, pp. 5610-5624, [DOI](#).
- **C. Schmidt** (2016) *High Resolution Integral-Field Spectroscopy of Gas and Ion Distributions in the Coma of Comet C/2012 S1 ISON*. *Icarus*, V265, pp. 35-41, [DOI](#).
- **C. Schmidt** (2013) *Monte-Carlo Modeling of North-South Asymmetries in Mercury's Sodium Exosphere*, *Journal of Geophysical Research*, V118, A50396, [DOI](#).

International Talks and Colloquia

Mercury's Sodium Exosphere, Mercury 2026, Leuven, BE	2026
Io's Atmosphere, Ionosphere, and Plasma Torus, LATMOS, FR	2023
The Lunar Sodium Exosphere, Taiwan Space Union, TW	2023
Solar Transit Spectroscopy of Mercury's Exosphere, Universiteit van Amsterdam, NL	2018
Io's Escaping Atmosphere & Plasma Torus, Universität zu Köln, DE	2018
Io's Plasma Torus Density & the S ⁺ Ribbon, Royal Institute of Technology, SE	2017
Small Telescope Applications: Mercury, Io & Comets, Université de Liège, BE	2017

Planetary Applications for Small Telescopes, Institute of Astronomy, Sofia, BG	2017
Visible Spectroscopy of the Io Plasma Torus, LESIA, l'Observatoire de Paris, FR	2016
Observations of Io, its Plasma Torus and Neutral Clouds, Lancaster Univ, UK	2016
Characteristics of Sodium Escape at Mercury, SERENA-HEWG, Killarney, IRL	2014

Active Grants & Fellowships (\$5.0M past, \$3.2M to Boston University)

- NASA/NExSCI Keck Award *Plasma drivers of Io's SO emission via joint Keck and JWST observations*, PI, 2026.02.12 to 2028.02.12. 7/2026A-N076. \$10,250.
- James Webb Space Telescope Cycle 4 *Io's Auroral Emissions as a Tool to Investigate Atmosphere-Plasma Torus Interactions*", Co-I (PI Zac Milby, Caltech), 2025.07.01 to 2028.06.30, JWST-GO-08857. \$110,475.
- Hubble Space Telescope Cycle 32 *A UV eclipse test for Callisto's atmosphere, plasma interaction and water ocean*, Co-I (PI Kurt Retherford, SwRI), 2025.03.01 to 2028.02.28. HST-GO-17739. \$63,963.
- NASA/NExSCI Keck Award *Joint Keck-Juno observations of Jupiter, its moons and its magnetosphere*, PI, 2024.08.01 to 2027.07.30. 80NSSC22K0954. \$120,000
- Hubble Space Telescope Cycle 31 *HST-Juno Io Campaign: Connecting Volcanos to the Plasma Environment*, Co-I (PI Kurt Retherford, SwRI), 2023.12.01 to 2026.11.30. HST-GO-17470. \$268,968.
- NSF AST PAARE *Accessing the dark Arizona skies for research and education, a NCAT-BU partnership*, Co-PI (PI Dan Clemens, BU), 2023.08.24-2026.08.23. \$256,696
- NASA FINESST PI for FI Emma Lovett *Characterizing and Simulating Alkalis in Europa's Exosphere*, PI, 2023.09.01 to 2026.08.31. 22-PLANET22-0198. \$150,000
- NASA Discovery Data Analysis Program *Mercury's escaping sodium tail*, PI, 2022.09.15 to 2027.09.14. 80NSSC22K1303. \$525,000
- NASA Discovery Data Analysis Program *Investigating the Impactor Contribution to Mercury's Exosphere*, Co-I/Institutional PI (PI Aimee Merkel, Univ. Colorado), 2021.05.21 to 2027.04.30. 80NSSC21K1019. \$131,866
- NASA Science Mission Directorate *Characterizing Mercury's Exosphere with BepiColombo-PHEBUS: US-based Co-Investigators*, PI, 2020.10.13 to 2026.10.12, 80NSSC21K0051. \$226,061

Teaching

- AS851 *Graduate Literature Seminar II* (10 students, Spring 2026)
- AS865 & AS866 *Graduate Research Seminar I & II* (~15 students. Fall & Spring 2022 - 2025)
- CC111 *Core Natural Science I: Origins- of the Big Bang, Earth, Life and Humanity* (96 students. Fall 2022)